Autonomous OS Debugging Agent

AI-Powered Real-Time Operating System Diagnostics, Remediation & Disaster Recovery

Project: Autonomous OS Debugging Agent	**Hackathon:** Build With Bharat 2.0
Author / Team: KernelHealers (Ansh & Team)	**Core Stack:** Python, Typer, Rich, PowerShell, OpenAI/Ollama

1. Explanation of the Project

The **Autonomous OS Debugging Agent** is an intelligent, terminal-based local systems engineering agent designed to diagnose, troubleshoot, and safely remediate complex operating system errors (such as Windows Update access denials 0x80070005, service startup faults, and corrupted component stores) without requiring manual research or dangerous trial-and-error commands.

Unlike generic LLM chatbots that produce blind advice without machine context, our agent acts as an autonomous Tier-3 Systems Administrator. It directly interfaces with kernel-level APIs and system loggers, executes non-destructive verification probes, generates production-grade PowerShell remediation scripts, and executes fixes under strict human supervision.

Key Core Objective: *Eliminate the friction and risks of OS troubleshooting by replacing cryptic error codes, unverified forum copy-pasting, and lengthy IT support tickets with an automated, self-healing diagnostic loop.*

2. How It Works (Technical Architecture & Pipeline)

The agent operates through a rigorous **5-step closed-loop reasoning pipeline**, ensuring that no modifying actions occur without prior telemetry verification and human consent:

- **Step 1: Privilege Verification & Environment Scaffolding:** On launch, the agent verifies elevation (Administrator via ctypes.windll.shell32.IsUserAnAdmin or POSIX root). It loads LLM configurations and dynamic system paths to ensure core Windows utilities (System32, PowerShell) are accessible.
- **Step 2: Live Log Ingestion & Telemetry Gathering:** The agent queries the Windows Event Viewer using PowerShell and WMI to extract the last 50 Critical (Level 1) and Error (Level 2) events across System, Application, and WindowsUpdateClient channels, packaging them into structured JSON.
- **Step 3: AI Diagnostic Engine & Safe Read-Only Probing:** An LLM reasoning loop formulates an initial hypothesis and outputs a suite of strictly read-only diagnostic commands (e.g. icacls, Get-Service, Get-ItemProperty). An execution sandbox validates commands against a destructive regex blacklist, executes them, and feeds stdout/stderr back to the LLM to confirm the true root cause.
- **Step 4: Fix Proposal & Human-in-the-Loop Approval:** Once the root cause is proven, the AI synthesizes a targeted PowerShell remediation script. The script is rendered in the terminal with Monokai syntax

highlighting and requires an affirmative [Y/n] confirmation before execution.

- **Step 5: Sandboxed Execution, Verification & Snapshot Rollback:** Before executing, a pre-fix snapshot and inverse rollback script are saved to .backups/<session_id>/. The fix runs via an isolated temporary file with guaranteed cleanup in finally: blocks. A final post-fix check confirms service restoration.

Pipeline Stage	Key Technologies / APIs	Safety Mechanism
Telemetry Ingestion	Get-WinEvent, WMI, platform	Read-only memory extraction
Hypothesis & Probing	OpenAI GPT-4o / Ollama Llama 3.1	Strict Regex Command Blacklist
Human Approval Gate	Typer Prompt, Rich Syntax Box	Mandatory explicit [Y/n] confirmation
Remediation & Rollback	Subprocess, WinReg, .backups/	1-Click instant state reversal

3. Why Is It Important? (Value & Enterprise Impact)

✓ **Drastic Reduction in Downtime (Hours to Seconds):** Traditional OS troubleshooting averages 45–90 minutes per incident involving manual log extraction, forum searches, and reboot cycles. The agent diagnoses and verifies solutions in under 15 seconds.

✓ **Zero-Hallucination Grounding:** Generic LLMs often hallucinate outdated or harmful registry commands because they lack real-time visibility into the machine. Our agent verifies actual filesystem ACLs and service statuses through live read-only probes before ever proposing a fix.

✓ **Enterprise Security & Air-Gapped Compliance:** The agent supports 100% offline execution via local Ollama models (e.g. Llama 3.1, Mistral). Defense, healthcare, and banking workstations with strict compliance policies can debug mission-critical machines without data leaving the subnet.

✓ **Deterministic Rollback (Zero-Risk Experimentation):** Every remediation automatically generates a pre-fix snapshot and an inverse rollback script. If a fix produces unintended side effects, running python agent.py rollback restores baseline system configurations in less than one second.

4. Offline Mode & Windows Recovery Environment (WinRE)

A. How to Trigger WinRE During a Boot Loop

- **The 3-Time Power Cut Safety Trigger:** Power on the machine; as soon as the manufacturer logo appears, press and hold Power for 8 seconds to force shut off. Repeat this 2–3 times. Windows automatically redirects into 'Preparing Automatic Repair'.
- **Manufacturer Hardware Hotkeys:** Press brand-specific recovery keys during power-on: HP (F11), Dell (F12/F8), Lenovo (F11 / Novo button), Asus (F9), Acer (Alt + F10).
- **Accessing the Command Prompt:** Inside the blue WinRE screen, navigate to: **Troubleshoot → Advanced options → Command Prompt** (which opens X:\windows\system32>cmd.exe).

B. Autonomous Offline Servicing Capabilities

- **Offline Event Log & Minidump Parsing:** Directly parses C:\Windows\System32\winevt\Logs\System.evtx and memory dumps (C:\Windows\Minidump*.dmp) to identify the crashing driver or faulting service.
- **Reverting Pending Update Actions:** Executes dism.exe /Image:C:\ /Cleanup-Image /RevertPendingActions to safely roll back corrupted Windows Updates that caused the crash.
- **Offline System Binary Verification:** Executes sfc.exe /scannow /offbootdir=C:\ /offwindir=C:\Windows to reconstruct missing system files.
- **Disabling Faulting Drivers Offline:** Loads and edits the offline registry hive (C:\Windows\System32\config\SYSTEM) to disable malfunctioning third-party driver services before the next boot.

Emergency Disaster Recovery Summary: *The agent is not restricted to healthy operating systems. Through WinRE and offline DISM servicing, it revives non-booting machines and prevents unnecessary hard drive reformatting.*